

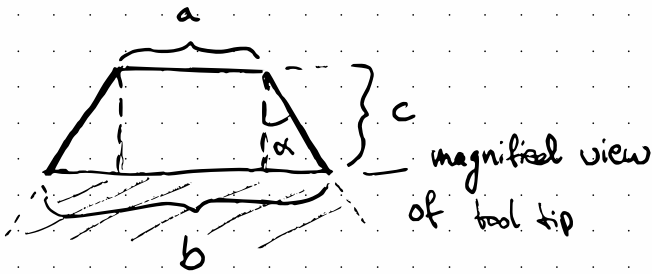
PCB tips & tricks

- Group Analog & Digital components to different sides of the board.



- All connectors on the same side and along one edge of the PCB.
- All SMD-components on the same side.
- Name all connector inputs and outputs!
- Mark all GND-pins with a small dot.
- Use a GND-plane on single sided boards.
- Use GND & power planes on dual sided boards.
- Give your board a name!
- Add M2-M3 mounting holes!
- Use a trace width calculator to make sure your PCB can handle the current.

Clearance & tolerances



a : tip diameter

α : tool angle / 2

h : cut depth

b : Isolation routing width.

length " b " is the smallest

possible for the specified tool and cut depth.

If you make smaller gaps than this then there will be short circuits in your PCBs!

Example

$$\begin{cases} \alpha = 30/2^\circ \\ a = 0,2 \text{ mm} \\ h = 0,1 \text{ mm} \end{cases}$$

What is b ? i.e. minimum clearance for our tool?

$$b = 2c + a$$

$$c = h \cdot \tan \alpha$$

$$b = a + 2h \cdot \tan \alpha$$

$$b = 0,2 + 2 \cdot 0,1 \cdot \tan 15^\circ = 0,2536 \text{ mm}$$

This means that the clearance should be

at least 0.2536 mm for a 30° tool at depth

- 0,1 mm

Rivets for Vias

inner diameter	outer dia
1	1.4
0.8	1
0.6	0.8
0.4	0.6

Use calculator tool in flatcam!

Preferably more!

Make sure that the vias in your design match the vias outer diameter. Otherwise

They won't fit!

Tools

Available tools in the workshop at C2

- Spiral bit, tip diameter: 0,2mm, angle: 60°
- Drill bit, 1mm
- Flat end milling bit, 1mm

Components

- Check the sizes of your components
- There are 0603 resistors and 0804 capacitors in the workshop. Use them!
- Smaller components than 0603 are very hard to place by hand!
- Double check that your footprints match your components!
- Double check that your drill holes matches your drills!
- If you order component samples from Würth Electronics. Send them to Martin Ekström.

FlatCam

settings

edit + preferences

MM or Inch depending on the units of the EDA

tool settings

Spindle speed: 11000 rpm
Feedrate x/y/z: 170 mm/min
travel z: 2 mm

} Maximum parameters of wegstn

One-Sided PCB

- Open copper layers and drill holes
- select All & move to origin if needed
(The wegstn will be angry otherwise)
- Calculate the depth of the v-bit (tools + Calculators)

Z: -0.1mm tool diameter 0.2mm tool angle 30°-60°
(see bit shaft)

- select the top layer from the working tree
 - Enter the calculated tool diameter and set passes: 3-4 passes
 - generate tool path
 - Enter rest of the tool settings
- 30% overlap

- Enter desired finishing position

Z: 35 mm

XY: (middle of workpiece, top+margin)
(this is just a convenience for inspection afterwards)

- Generate CNC-code and save as .nc or .gcode

Drilling - drill bit

- Use 1mm drill bit. Make sure that all drilled holes are 1mm wide!
- select the Plated through hole (PTH) file in Flatcam and go to the drilling tool.
- Set the depth to -1.65 mm and tool diameter to 1mm.
- generate geometry & G-code

Mill holes

- select the file with the larger than 1mm holes. mark the holes to mill, then press "Utilities" & mill holes. set the tool width to 1mm and depth to -1.65 mm. Check "multiple passes" box and set the value to 0.55 mm.
- Generate geometry & G-code

Edge cuts

- Select the "board outline" file or "edge cut"
- Use the "cutout tool", set the same parameters as when milling holes. i.e. -1.65mm deep, multiple passes at 0.55mm & 1mm tool diameter.
- Set the "bridge gap" parameter to 0.5-1mm. And decide where your gaps will be. "4" or "2TB" works well.

Use the vacuum as you cut the board shape. FR4 is fibreglass and you don't want that in your lungs!

Wegstr

Copper Laminate

100 x 160 mm

workspace size

140 x 200 mm

- Vacuum after each CNC job is done!
- fasten the copper plate with at least 4 clamps
- import .nc or .gcode files
- Position the spindle at a position where the machine can run its path without any collisions. Also try to minimize waste. Don't put small pieces of pcb in the center of a copper board.
- Make sure the autolevel clamps are attached and lower the spindle until it stops.
- Set all the axes to zero and run autolevel procedure. this takes a few minutes depending on the size of the pcb.
- Run CNC Job. Wegstr will prompt for tool changes. Don't change tool for first prompt since it is already calibrated for this.

Tool Change

When you are doing a tool change after autolevel you shall only reset the Z-axis, the same way as before. Do not rerun autolevel, unless you move your design!

Cut-out PCB

Use the cut out tool in Flatcam